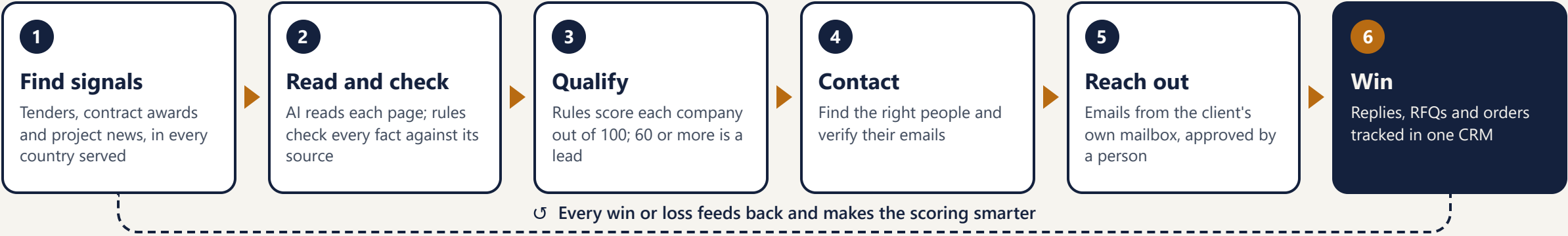


# AI lead generation for pipeline and industrial projects

Assessment and build proposal · September 2026 · One-page summary; full detail and diagrams follow



**Optional premium services** can plug into steps 1, 3, 4 and 5 (project databases, trade records, contact data, sending tools) for even better leads.

**6 work packages**

Security first, then discovery, contacts, validation, outreach and handover

**100-point score**

Seven source criteria and clear rules decide who is a customer

**\$130–365 / month**

Running cost once live: Claude API usage (~\$100–300), database and server. AI coding tools for the build are separate.

**85% → 90%**

Leads correct: foundation stage → after tuning

**Where the prototype stands**

- Covers about a quarter of the target workflow
- Open database and no login: fixed first, in work package 0
- The good parts (interface, evidence rules) are kept

**How we build it**

- Free public sources in every target country, plus our own code
- Seven source criteria and a 100-point score decide who is a customer
- Claude AI reads; rules decide; every fact is linked to its source

**What the client gets**

- Qualified buyers with evidence, in any market
- Verified contacts and outreach from its own mailbox
- One CRM from first signal to order; it owns the data and code

# Assessment and build proposal

Full detail · September 2026 · Diagrams appear on landscape pages next to the sections they belong to.

## Summary

1. The target workflow
2. Where the prototype stands today
3. Issues that block real use
4. Positioning: procurement intent, not web-browsing intent
5. Free first: build in-house and use what the client already has
6. How we decide who is a customer
7. How we build it in-house: rules decide, AI reads

8. Beyond contract awards: the full buyer funnel
  9. Target architecture: an in-house harness with premium plug-ins
  10. Roadmap: work packages
  11. How it changes over time, and benchmarks
  12. Is it practical?
  13. Development requirements
  14. Development and running cost: tools and APIs only
- Sources

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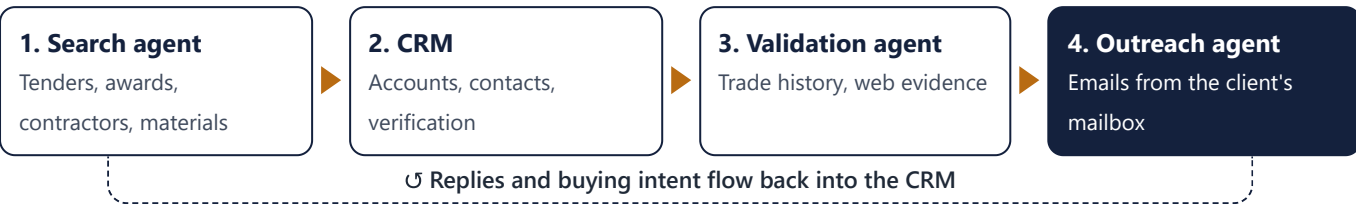
## Summary

We recommend a build in six work packages that turns the current prototype into an AI system that finds buyers for pipeline and industrial projects in every country the client serves, including the GCC, from public procurement signals, confirms their need and opens conversations from the client's own mailbox. It runs on free sources, open-source software we host ourselves, and tools the client already has. The only recurring cost is AI usage and hosting, about \$130–365 a month.

- **Where it stands:** the prototype is well organised and covers about a quarter of the target workflow. Search partly works, the CRM is mostly screens, buyer validation is a placeholder and outreach does not exist.
- **What must change first:** the database is open to anyone with the URL, there is no login, contact details can be attached to the wrong person, and the dashboard shows sample data. The first work package fixes all of this.
- **Free first:** search, page reading, contacts, background jobs and the CRM run on free sources, self-hosted open-source tools or the client's existing tools. Paid data is optional and is added only if a measured gap costs deals (section 5).
- **How we decide who is a customer:** seven criteria choose which websites to watch, and a 100-point score, calculated by rules, decides which companies qualify (section 6).
- **Rules decide, AI reads:** deterministic code collects, checks, matches, scores and enforces sending limits. Claude only reads text and writes, and code checks every AI answer (section 7).
- **More than contracts:** eleven buying signals, six buyer types and an eight-stage funnel from first signal to purchase order (section 8).
- **A harness with premium plug-ins:** every step has a free in-house default and an optional premium provider behind the same interface, so the client can buy top-notch quality where it pays off without rebuilding anything (section 9).
- **Six work packages:** from securing the prototype to handover, each with a measurable acceptance test; the timeline is agreed with the client (section 10).
- **Benchmarks:** 85% or more of leads correct at the foundation stage and 90% after tuning, a reply rate at or above the 3.43% cold-email average, and a spam rate under Google's 0.3% limit (section 11).
- **Development requirements:** nine specialities covering AI, data, product, email deliverability, security, operations, QA, industry knowledge and design (section 13).
- **Cost:** about \$130–365 a month to run, plus AI coding tools (Claude Code, Codex) at \$20–200 per developer per month during the build (section 14).

# 1. The target workflow

The client wants an AI sales team that finds industrial buyers before competitors do, proves they need the product, and starts the conversation from the client's own mailbox. There are four stages, and every stage writes back to one CRM.



The loop back from stage 4 matters most: replies are the strongest intent signal the system will ever get.

| Stage               | What it must do                                                                                                                                                       | Output into the CRM                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1. Search agent     | Take a topic or project, find who won the tender, the EPC contractor and subcontractors, their key people, and the materials the project needs (pipes, steel, valves) | Project, owner, contractors, material packages, contacts, source evidence             |
| 2. CRM              | Show everything found, and verify contact details                                                                                                                     | Verified contacts, stage, owner, activity history                                     |
| 3. Validation agent | Research each buyer's past purchases on the web and in Volza trade data; confirm they really need the suggested products                                              | Need score with reasons: "imported 4,200 t of API 5L line pipe in the last 12 months" |
| 4. Outreach agent   | Send proposals from the client's mailbox, handle replies, test buying intent                                                                                          | Emails sent, replies, intent status updated automatically                             |

# 2. Where the prototype stands today

The prototype covers roughly a quarter of the target workflow. Stage 1 works partly, stage 2 mostly as screens, stage 3 is a placeholder and stage 4 does not exist. We audited the full codebase (about 15,000 lines) on 24 Sep 2026: it type-checks cleanly, all 51 automated tests pass and the production build succeeds.

| Stage               | Built | What exists                                                                                                                                                                 | What is missing                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Search agent     | ~55%  | Live search (Google via SerpAPI), page scraping (Firecrawl), AI extraction of company, project, contractor and material fields, source-quality filters, five search "plays" | It is a fixed pipeline of about 2,000 lines of keyword rules, not an agent. Each run lives inside one web request capped at 28 seconds, so only 6–12 pages are read. No scheduled watchlists, no tender-portal connectors and no local-language sources. Parts are hard-coded to a few sample queries and one region's buyer list, so it cannot be pointed at other countries without code changes. |
| 2. CRM              | ~40%  | Screens for candidates, leads, decision makers, lead lists, activities and CSV export; 13 database migrations                                                               | No login. Contact enrichment and email verification queues fill up but nothing processes them, so no email can ever reach "Verified". Data is split between the browser and the database. The dashboard shows hardcoded sample records, not real data.                                                                                                                                              |
| 3. Validation agent | ~5%   | A trade-data interface with a Volza placeholder that always returns "Pending"                                                                                               | Volza integration, purchase-history matching, need scoring, the agent itself                                                                                                                                                                                                                                                                                                                        |
| 4. Outreach agent   | 0%    | Nothing                                                                                                                                                                     | Mailbox connection, sending-domain setup, sequences, AI drafting, reply handling, CRM auto-update, opt-out handling                                                                                                                                                                                                                                                                                 |

**What the FDE did well:** clean module boundaries with swappable providers, AI output validated before use, an evidence-first rule (no invented emails, unknown stays unknown), transparent run logs showing why each source

was accepted or skipped, and honest client documentation of what is planned versus built. We keep all of this. What we replace is the engine underneath it and everything that is missing.

### 3. Issues that block real use

Three issues make the prototype unsafe to put in front of a sales team today: the database is open to anyone, contact data can be wrongly attached to people, and the dashboard is not real. All of them are fixed in the first work package.

| # | Issue                                                            | Evidence in the code                                                                                                                                                                                         | Impact                                                                                                              | Severity |
|---|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------|
| 1 | Anyone with the app URL can read, change or delete the whole CRM | <code>/api/discovery/crm</code> uses the database's master (service-role) key, which bypasses all access rules, and checks no login. <code>proxy.ts</code> enforces nothing and excludes <code>/api</code> . | Full leak of leads, contacts and notes; data can be wiped                                                           | Critical |
| 2 | No login                                                         | <code>/login</code> is a placeholder: "Auth wiring pending credentials"                                                                                                                                      | No users, roles or audit trail                                                                                      | Critical |
| 3 | Master database key was shared in chat                           | Stated in the project's own setup notes                                                                                                                                                                      | Must be rotated before any deployment                                                                               | Critical |
| 4 | Contacts can get another person's LinkedIn profile or email      | <code>nearestLinkedIn()</code> falls back to the first LinkedIn URL on the page; <code>nearestEmail()</code> gives a page's only email (often info@) to every person and labels it "source-backed"           | Wrong person contacted; damages sender reputation and trust                                                         | High     |
| 5 | Dashboard shows hardcoded sample data                            | <code>repository.ts</code> returns records from <code>verified-data.ts</code> and never reads the database                                                                                                   | Management sees numbers that don't reflect real work                                                                | High     |
| 6 | Two competing data models                                        | Discovery writes flat <code>discovery_*</code> tables with text IDs; the structured company, signal and score tables are never used                                                                          | No de-duplication; no project → owner → contractor → material graph                                                 | High     |
| 7 | Buyer score is mostly fixed                                      | Every company gets +5 "company fit" automatically; intent and relationship scores are always 0                                                                                                               | Ranking is not trustworthy                                                                                          | Medium   |
| 8 | Personal data collected without a compliance basis               | Named people are pulled from Google results and LinkedIn snippets and stored with no consent, retention or opt-out handling                                                                                  | Exposure under data-protection laws in each market (for example UAE PDPL, Saudi PDPL, EU GDPR) and LinkedIn's terms | Medium   |
| 9 | Unreproducible builds                                            | Almost every dependency is pinned to <code>latest</code>                                                                                                                                                     | An upstream release can break the build without warning                                                             | Low      |

### 4. Positioning: procurement intent, not web-browsing intent

We recommend not cloning 6sense, Bombora or ZoomInfo. Their "intent" is built for software buyers researching online, and it barely sees an EPC contractor buying line pipe for a pipeline project. For industrial project buyers, buying intent is public and specific: a tender, a contract award, an import record, a reply to an email.

|                                   | 6sense / Bombora / ZoomInfo                                                                                                                      | This platform                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| What "intent" means               | An anonymous company reading more articles on a topic (Bombora's publisher network), website visits (6sense), plus a contact database (ZoomInfo) | A named project needs a named material, won by a named contractor, within a procurement window |
| Signal source                     | Mostly English-language business and tech websites                                                                                               | Tender portals, EPC award news, owner announcements, import and export records, email replies  |
| Fit for industrial project buyers | Weak: procurement happens through portals and relationships, not web research                                                                    | Strong: built around how industrial projects are procured, country by country                  |

|              | 6sense / Bombora / ZoomInfo                                                    | This platform                                                         |
|--------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Evidence     | A topic score with no source                                                   | Every claim links to a source page and the exact quote                |
| Typical cost | Enterprise contracts, commonly tens of thousands of dollars a year per product | Mostly free public sources; we pay only for AI usage (see section 14) |

The FDE's roadmap suggested adding all 12 layers of the 6sense stack, including technographics and website-visitor identification. For a steel and pipe supplier, those layers add cost without adding buyers. We keep two ideas from those products: an explainable account score, and a contact lookup that tries several sources in turn.

## 5. Free first: build in-house and use what the client already has

Almost every layer can run on free sources, open-source software we host ourselves, or tools the client already pays for. The only unavoidable usage cost is the AI model. Paid data stays optional and is added only if a measured gap is costing deals.

| Layer                        | Free or in-house way                                                                                                                                                                                                             | What the client may already have | Paid option, only if needed later                               |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------|
| Finding news and projects    | <a href="#">GDELT</a> news API (free, no key), Google News and trade-press RSS feeds, owner newsrooms, <a href="#">Brave Search API</a> within its \$5 monthly free credit (~1,000 searches)                                     | None                             | Brave or Claude web search at \$5–10 per 1,000 searches         |
| Reading web pages            | Self-hosted <a href="#">Crawl4AI</a> or <a href="#">Crawlee</a> (both Apache 2.0) on our own server; <a href="#">Jina Reader</a> free key                                                                                        | None                             | Firecrawl cloud                                                 |
| Tender portals               | Our own reader for each portal                                                                                                                                                                                                   | Supplier registrations           | Tender booklets, bought only for priority tenders               |
| Stock-exchange announcements | Our own reader for exchange announcement pages in each target market                                                                                                                                                             | None                             | None                                                            |
| Contacts                     | Company websites, press releases and annual reports; email format worked out from public addresses; self-hosted <a href="#">Reacher</a> mail-server check; <a href="#">Hunter</a> free plan (50 credits a month) for spot checks | None                             | Apollo from ~\$49 per user a month, if the fill rate is too low |
| Import history               | <a href="#">UN Comtrade</a> (free, country level) for market sizing                                                                                                                                                              | Volza account                    | Volza, if not already held                                      |
| CRM                          | <a href="#">Zoho CRM Free</a> (up to 3 users) or our own CRM screens                                                                                                                                                             | Existing CRM                     | None needed                                                     |
| Email sending                | Gmail or Microsoft 365 API                                                                                                                                                                                                       | Existing company email           | Extra mailboxes on a sending domain                             |
| Background jobs              | A Postgres job queue ( <a href="#">Supabase Queues</a> and pg_cron)                                                                                                                                                              | None                             | Inngest or Trigger.dev                                          |
| Sanctions screening          | Free <a href="#">OFAC</a> , <a href="#">UN</a> and <a href="#">UAE</a> lists                                                                                                                                                     | None                             | None                                                            |
| AI                           | Claude API, pay per use; the <a href="#">Batch API</a> halves the price of scheduled runs                                                                                                                                        | None                             | None                                                            |

Two free options we deliberately avoid. Google's Custom Search API is [closed to new customers](#) and shuts down on 1 Jan 2027. The free Gemini API tier allows Google to [use prompts to improve its products](#) for UAE users, so we don't send client data there.

### Can we get intent data from CDN or internet providers?

No. CDNs only process traffic for their own customers and state that they do not use or sell end-user data for marketing ([AWS](#), [Cloudflare](#), [Fastly](#)). Buying it would also breach data-protection law: the [UAE PDPL](#) treats online

identifiers such as IP addresses as personal data, with consent as the default basis, and the [Saudi PDPL](#) requires consent or prior interaction for direct marketing. The EU GDPR also treats IP addresses as personal data.

The established vendors get intent data like this:

- **Bombora:** a [co-op](#) of thousands of B2B websites that run Bombora's tag and share consented reading data.
- **6sense:** its own network of B2B web activity, keyword tracking in more than 40 languages, and partner feeds including Bombora, G2 and TrustRadius ([6sense](#)).
- **ZoomInfo:** a contact database built over many years from licensed, contributed and researched data.

For this client, the detail comes from better sources:

- **Tender documents.** Many government portals share or sell full tender documents containing the bills of quantities, with pipe sizes, grades and quantities. Saudi Etimad's rules are one example ([executive regulations](#), articles 33 and 69).
- **Import records.** Volza shows each shipment's product code, quantity, supplier and date.
- **Award chains.** An EPC contract award is usually followed, within weeks to a few months, by bulk orders for line pipe, plates and valves. The platform tracks that chain from award to order.
- **The client's own signals.** Website visits, email opens, clicks and replies, RFQ forms, and meetings at industry trade shows in each market.

## 6. How we decide who is a customer

A company becomes a lead only when a written checklist says so. First we choose which websites are worth watching, using seven criteria. Then every company found there is scored against the client's customer rules. The rules are plain, visible and editable by the sales team.

### Which websites we watch

| Criterion  | Rule                                                                                                           |
|------------|----------------------------------------------------------------------------------------------------------------|
| Relevant   | Publishes projects, tenders, contract awards or procurement news in the client's sectors and target countries  |
| First-hand | Official or primary sources (government portals, project owners, listed companies) rank above republished news |
| Fresh      | Updates at least weekly                                                                                        |
| Open       | Public page, RSS feed or API; respects robots.txt and the site's terms; never bypasses a login                 |
| Readable   | Stable HTML, RSS or PDF that we can parse reliably                                                             |
| Language   | English and the local languages we support                                                                     |
| Proven     | Keeps its place only if its items turn into qualified leads, measured monthly                                  |

| Source category              | Examples                                                                                                                                   | How we read it                        | Tier |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------|
| Government tender portals    | The national e-procurement portal of each target country, e.g. Saudi Etimad, EU TED, US SAM.gov, UK Find a Tender; several offer free APIs | A reader built for each portal, daily | A    |
| Project owners' newsrooms    | Oil, gas, water and power companies, pipeline operators and utilities in each market                                                       | RSS or page crawl, daily              | A    |
| Stock-exchange announcements | Contract awards announced by listed owners and contractors, e.g. Saudi Tadawul, UK regulatory news, US SEC filings                         | Announcement feeds, daily             | A    |
| Trade press                  | Pipeline, oil and gas, water and construction trade publications                                                                           | RSS headlines, then the article       | B    |
| Global news index            | GDELT                                                                                                                                      | Free API, several times a day         | B    |

| Source category                       | Examples                                              | How we read it | Tier |
|---------------------------------------|-------------------------------------------------------|----------------|------|
| Contractor websites and careers pages | EPCs and fabricators already in the buyer graph       | Crawl weekly   | B    |
| Trade records                         | Volza (client's account), UN Comtrade (country level) | Export or API  | A    |

The source list is set up per country. Adding a new market means adding its portals, exchanges and news sources to the list, not changing code.

### Which companies count as customers (score out of 100)

| Rule        | What it checks                                                                                             | Points |
|-------------|------------------------------------------------------------------------------------------------------------|--------|
| Product fit | The project or company needs a product family the client sells, matched by keywords and customs (HS) codes | 0–25   |
| Timing      | The signal is inside the buying window: open tender, award in the last 9 months, RFQ                       | 0–20   |
| Buying role | Owner, EPC contractor, subcontractor, fabricator or stockist, weighted by how often each buys directly     | 0–15   |
| Market      | Located in, or delivering to, the countries the client serves                                              | 0–10   |
| Size        | Project value or yearly import volume above a threshold the client sets                                    | 0–10   |
| Evidence    | Official source, exact quote stored, confirmed by a second source                                          | 0–10   |
| Reachable   | At least one verified decision maker                                                                       | 0–10   |

**Hard stops (never scored):** existing customers (sent to their account owner), competitors, sanctioned companies (checked against the free UN and US OFAC lists), completed or cancelled projects, and markets the client doesn't serve.

A company scoring 60 or more becomes a qualified lead. The threshold and weights start as agreed defaults and are tuned from the client's won and lost deals. AI reads the pages and fills in the facts, but the score is always calculated by these rules, so the same facts always give the same score.

## 7. How we build it in-house: rules decide, AI reads

Every step runs deterministic code first and uses AI only where language has to be understood. Code is free to run, repeatable and auditable. AI is paid per use, so it is reserved for reading unstructured text and writing, and code checks every AI answer before it is saved.

| Step               | Deterministic code and rules                                                                                     | AI (Claude)                                                                                           | Free tools                         |
|--------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------|
| 1. Collect         | Source list, schedules, RSS and API readers, portal readers, de-duplication by URL and content fingerprint       | None                                                                                                  | Crawl4AI or Playwright, feedparser |
| 2. Filter          | Keyword, region, date and source-tier rules drop most pages                                                      | A low-cost model (Claude Haiku) only for pages the rules can't decide                                 | None                               |
| 3. Extract         | Parse dates, values, reference numbers, tables and PDF text                                                      | Reads articles and tender text to pull out companies, roles, scope and materials, each with its quote | PDF text extractors                |
| 4. Check           | The quote must appear word-for-word on the page; values must be from allowed lists; dates and numbers must parse | None                                                                                                  | Schema validation                  |
| 5. Match companies | Name normalisation, alias table, fuzzy matching in Postgres, website-domain match                                | Tie-breaker for unclear names, e.g. the same company written in two languages                         | Postgres trigram search            |

| Step             | Deterministic code and rules                                                                                                  | AI (Claude)                                              | Free tools                            |
|------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------|
| 6. Score         | The customer rules in section 6                                                                                               | Writes the plain-language reason                         | None                                  |
| 7. Find contacts | Crawl leadership pages, team pages and press releases; work out the email format from public addresses; check the mail server | Maps job titles to buying roles                          | Reacher (self-hosted email checker)   |
| 8. Validate need | Match customs codes in import records; count past awards                                                                      | Summarises the evidence for the salesperson              | Volza (client's account), UN Comtrade |
| 9. Outreach      | Templates, daily sending limits, opt-out and suppression list, send windows                                                   | Drafts the personal opening lines; sorts replies         | Gmail or Microsoft 365 API            |
| 10. Learn        | Conversion rate per source and per signal; rule weights adjusted                                                              | Suggests new rules from patterns; a person approves them | None                                  |

Over time the split moves further towards code. When the AI keeps making the same judgement, we turn it into a rule. That lowers the cost per lead and makes results more consistent.

## 8. Beyond contract awards: the full buyer funnel

Contract awards are one buying signal out of about ten. The platform follows a project from its first budget line to the material order, finds buyers who never appear in a tender, and moves every account through an eight-stage funnel.

### Buying signals, earliest first

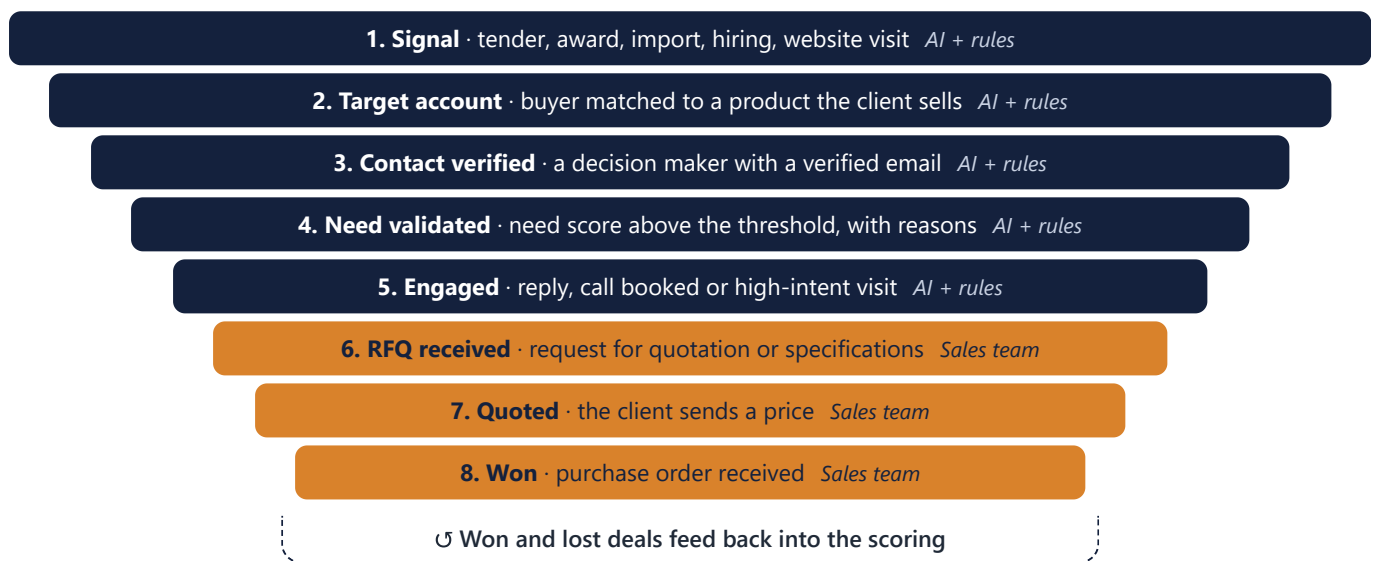
| Signal                                              | Example                                                                 | Source                                   | Approximate time before material purchase |
|-----------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------|-------------------------------------------|
| Capex plan or budget                                | An owner adds a pipeline expansion to its five-year plan                | Annual reports, government budgets, news | 18–36 months                              |
| Project announced or design (FEED) contract awarded | Front-end engineering design contract signed                            | News, project databases                  | 12–24 months                              |
| Permits and approvals                               | Environmental approval or land allocation granted                       | Regulator notices                        | 12–18 months                              |
| Tender or prequalification                          | EPC tender issued; bidders prequalified                                 | Tender portals                           | 6–12 months                               |
| EPC contract award                                  | Owner names the winning EPC contractor                                  | Press releases, portals                  | 1–9 months (peak window)                  |
| Subcontract or package award                        | Piping or structural package let to a subcontractor                     | News, contractor announcements           | 1–6 months                                |
| Hiring                                              | Contractor hires piping or procurement staff for the project            | Job boards                               | 3–9 months                                |
| Material RFQ or vendor registration                 | EPC opens vendor registration or sends RFQs                             | Contractor vendor portals, email         | 0–3 months (hot)                          |
| Import history                                      | A company imported line pipe every quarter last year                    | Customs records (Volza)                  | Recurring                                 |
| Maintenance shutdown                                | A refinery schedules a turnaround                                       | Operator notices, news                   | 3–6 months                                |
| Engagement                                          | Visits the client's website, replies to an email, meets at a trade show | Client's own website, mailbox and CRM    | Now                                       |

Timings are indicative and vary with project size. The earlier the signal, the more time to get onto the approved vendor list before the order is placed.

## Buyers who are not contract winners

| Buyer type                                         | What they buy                                             | How the platform finds them               |
|----------------------------------------------------|-----------------------------------------------------------|-------------------------------------------|
| Project owners (national oil companies, utilities) | Materials they procure directly and supply to contractors | Capex plans, tenders                      |
| EPC contractors                                    | Bulk materials for awarded projects                       | Contract awards                           |
| Subcontractors and fabricators                     | Piping spools, structural steel, packages                 | Package awards, hiring                    |
| Stockists, distributors, traders                   | Regular restocking for resale                             | Import records                            |
| Operating plants (maintenance)                     | Spares for shutdowns and repairs                          | Shutdown schedules, annual rate contracts |
| Equipment makers                                   | Plate and pipe for tanks, skids, pressure vessels         | Import records, product catalogues        |

## The funnel



The agents run stages 1–5; the sales team owns 6–8, and every won or lost deal feeds back into scoring.

| Stage               | Moves forward when                                                        | Done by                                     |
|---------------------|---------------------------------------------------------------------------|---------------------------------------------|
| 1. Signal           | A buying signal from the table above is captured with its source          | Discovery agent                             |
| 2. Target account   | The buyer company is identified and matched to a product the client sells | Discovery agent                             |
| 3. Contact verified | At least one decision maker has a verified email                          | In-house contact finder                     |
| 4. Need validated   | Need score passes the threshold, with written reasons                     | Validation agent                            |
| 5. Engaged          | The buyer replies, books a call or visits a high-intent page              | Outreach agent                              |
| 6. RFQ received     | The buyer sends a request for quotation or specifications                 | Sales team, logged automatically from email |
| 7. Quoted           | The client sends a price                                                  | Sales team                                  |
| 8. Won              | Purchase order received                                                   | Sales team                                  |

The dashboard shows conversion between every stage and which signal types produce orders, so budget moves to the signals that work.

## 9. Target architecture: an in-house harness with premium plug-ins

The platform is a harness. Our own code runs every step, each step has a free in-house default, and each step can take an optional premium provider through the same interface. The in-house version is good on its own; premium plug-ins are how the client buys top-notch lead quality in specific places, only when a measured gap justifies it.

The three diagrams at the end of this section show how the platform works, the system architecture, and the free version against real premium services.

| Layer                   | Choice                                                                                                                                            | Why                                                                                                      |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| App and UI              | Keep Next.js, React, TypeScript, Tailwind                                                                                                         | The prototype's UI is solid; rebuilding it wastes time                                                   |
| Database                | Keep Supabase Postgres; one schema; row-level security; trigram search for matching company names                                                 | Already in place; the unused structured tables from migration 0001 become the base of the graph          |
| Background jobs         | A Postgres job queue (Supabase Queues and pg_cron)                                                                                                | Durable multi-step runs, retries and schedules with no extra paid service                                |
| AI models               | Claude Haiku 4.5 for filtering, Claude Sonnet 5 for extraction, Claude Opus 5 for qualification                                                   | Strong structured output; used only where rules can't do the job; the provider interface stays swappable |
| Search and page reading | GDELT, RSS feeds and self-hosted Crawl4AI behind interfaces; our own readers for tender portals and exchange announcements in each target country | Free, and the portals are higher signal than general web search                                          |
| Contacts                | In-house: company sites and press releases, email format from public addresses, self-hosted mail-server check                                     | Free and owned; a paid source is added only if a measured gap costs deals                                |
| Trade data              | The client's Volza account, if they have one; UN Comtrade for country totals                                                                      | Past imports by customs code prove a buyer uses the product                                              |
| Email                   | Microsoft 365 or Google Workspace API with OAuth, on a separate sending domain                                                                    | Emails come from real mailboxes with proper authentication                                               |
| Observability           | Sentry free tier for errors, a job dashboard, cost logged per run                                                                                 | Know what failed and what each lead cost                                                                 |

### In-house versus premium plug-ins, layer by layer

| Layer            | In-house default                                         | Premium plug-in                                      | Quality uplift                                                           | Cost band   | Switch on when                                      |
|------------------|----------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------|-------------|-----------------------------------------------------|
| Signal discovery | GDELT, RSS, newsroom, portal and exchange readers        | Paid project databases covering the client's markets | Earlier stages (concept, design), subcontract packages, project contacts | Medium      | Too few early-stage projects in the pipeline        |
| Page reading     | Crawl4AI on our own server                               | Firecrawl cloud                                      | JS-heavy and bot-protected sites; less maintenance                       | Low         | More than 10% of priority sources fail to read      |
| Material detail  | Public tender notices                                    | Paid tender booklets                                 | Exact sizes, grades and tonnage                                          | Per tender  | A priority tender in the client's product range     |
| Company data     | Own buyer graph and sanctions lists                      | Apollo or ZoomInfo company data                      | Size, parent company, web domains, industry codes                        | Low to high | Missing domains or bad matches block contact search |
| Contacts         | Websites, press releases, email format and Reacher check | Apollo, then Lusha, Cognism or ZoomInfo; ZeroBounce  | More mid-level buyers, direct dials, verified emails                     | Low to high | Verified contact on under 50% of priority accounts  |

| Layer           | In-house default                             | Premium plug-in                               | Quality uplift                                                                | Cost band | Switch on when                                        |
|-----------------|----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------|-----------|-------------------------------------------------------|
| Need validation | Client's Volza, UN Comtrade, past awards     | Volza full plan or API                        | Company-level import history per customs code                                 | Medium    | Qualified leads are not turning into RFQs             |
| Intent timing   | Own procurement signals and engagement       | Bombora or 6sense                             | Research-surge timing (thin coverage for industrial buyers in many countries) | High      | Only after a coverage test on the top 200 accounts    |
| Outreach        | Gmail or Microsoft 365 API, limits, approval | Instantly or Smartlead, extra sending domains | Warm-up, inbox rotation, deliverability at scale                              | Low       | Sending more than about 50 emails per mailbox per day |

Cost bands: low is under 100 US dollars a month, medium is 100–1,000, high is over 1,000 (checked September 2026). Each plug-in is trialled for 4–6 weeks on the same accounts and kept only if qualified leads, replies or RFQs rise. Suggested order: email verification, contact database, Volza, project database, then intent data.

#### How the agents avoid making things up:

- **Quote-backed facts.** Every extracted field stores the exact source sentence. A field with no quote is not saved.
- **A test set of real awards.** About 150 known contract awards and tenders from the client's markets with verified answers. Every change to prompts or code is scored against it before release.
- **Merging duplicates.** "Acme Pipelines Ltd", "Acme Pipeline Co." and the same name in a local script resolve to one company record.
- **People approve before sending.** In the first release the outreach agent drafts and a person approves; auto-send is unlocked per campaign once quality is proven.
- **Spending limits.** Each run has a cap on searches, pages read and AI spend.

# How the platform works

One platform, five steps. It works on free sources, and premium data can be plugged in for even better leads.



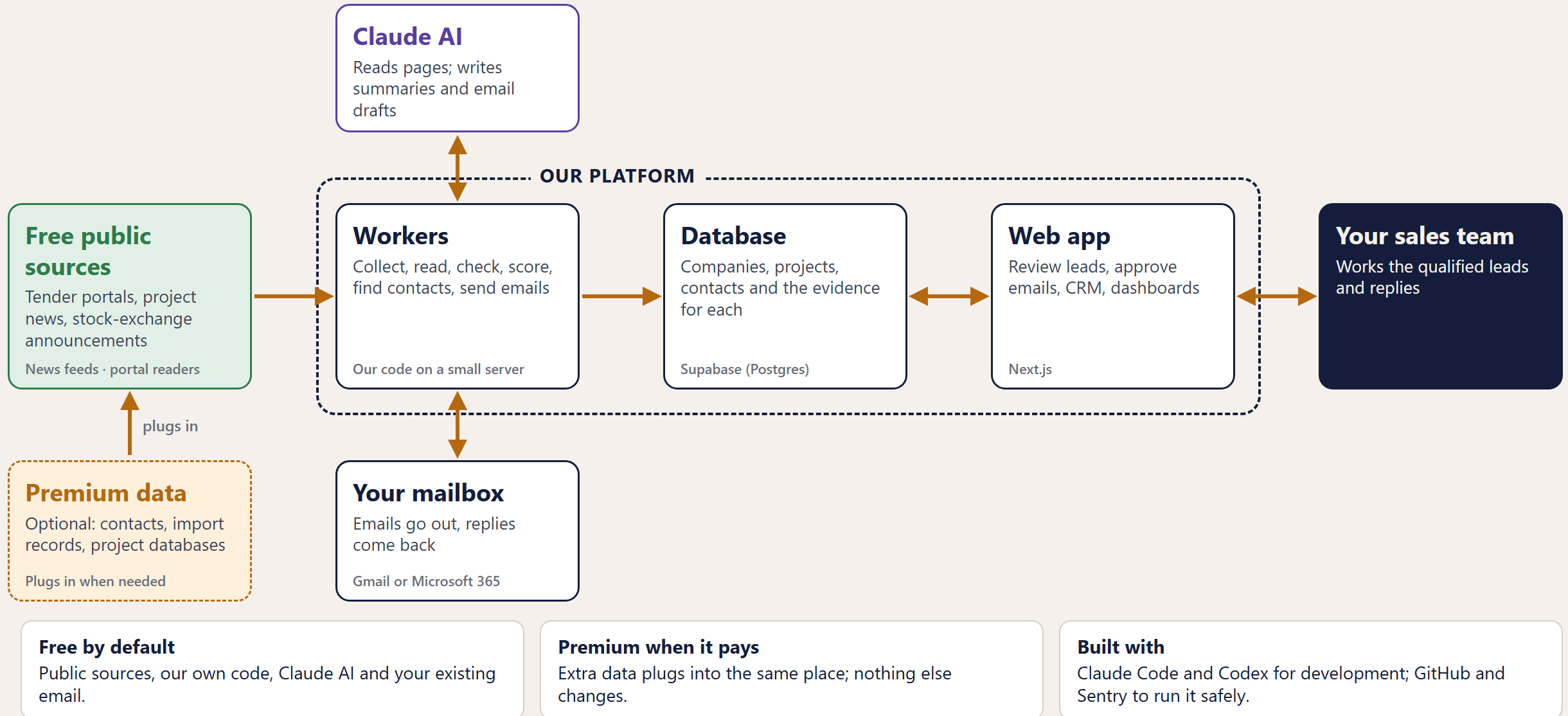
**Free version**  
Good leads from public sources. Runs for about \$130–365 a month.

**Premium add-ons**  
Better leads where public data is thin. Switch on one at a time.

**Same platform either way**  
Adding premium data is a setting, not a rebuild.

# System architecture

What we build and run (inside the dashed line), and the outside services it connects to.



# Free version versus premium services

Every step works for free. These real services can be added, one at a time, to raise lead quality where it pays off.

| Step                      | Free (built in)                                                                     | Premium services you can add                                                                                                                                      | What you gain                                                                                                                                  | Cost        |
|---------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Find projects and tenders | News feeds, government tender portals, company and stock-exchange announcements     | <b>GlobalData</b> , <b>IJGlobal</b> , <b>Rystad Energy</b> (project databases) · <b>Tendersinfo</b> , <b>GlobalTenders</b> (tender aggregators)                   | Projects found months earlier, at planning and design stage; named contractors and subcontractors; project values; one feed for many countries | \$\$-\$\$\$ |
| Read hard websites        | Our own page reader (Crawl4AI) + Claude AI                                          | <b>Firecrawl</b> , <b>Zyte</b> , <b>Bright Data</b> , <b>Apify</b>                                                                                                | Pages behind JavaScript or bot protection read reliably; fewer failed sources; less upkeep                                                     | \$-\$\$     |
| Prove they buy            | UN Comtrade country totals, past contract awards                                    | <b>Volza</b> , <b>ImportGenius</b> , <b>Panjiva</b> (S&P Global) trade records                                                                                    | Each company's actual shipments by product code: volumes, prices and current suppliers. Real buyers, their size, and who you would replace     | \$-\$\$\$   |
| Know the company          | Our own records, public filings, free sanctions lists                               | <b>Dun &amp; Bradstreet</b> , <b>Orbis</b> (Moody's)                                                                                                              | Size, revenue, parent and subsidiaries, credit risk: target the entity that decides, avoid slow payers                                         | \$\$\$      |
| Spot the timing           | Tender and award dates; visits and replies from the client's own website and emails | <b>TheirStack</b> , <b>Coresignal</b> (hiring data) · <b>Bombora</b> , <b>6sense</b> (intent data)                                                                | Hiring for procurement or project roles shows a project starting; intent shows research surges (thin for industrial buyers, so test first)     | \$-\$\$\$   |
| Find and verify contacts  | Company websites, press releases, email-format check (Reacher)                      | <b>Apollo</b> , <b>Lusha</b> , <b>Cognism</b> , <b>ZoomInfo</b> · <b>Clay</b> or <b>FullEnrich</b> (search many at once) · <b>ZeroBounce</b> , <b>NeverBounce</b> | More decision makers, including mid-level buyers; direct phone numbers; every email verified, so bounces stay under 2%                         | \$-\$\$\$   |
| Reach out at scale        | The client's own mailbox (Gmail or Microsoft 365), Google Postmaster Tools          | <b>Instantly</b> , <b>Smartlead</b> , <b>lemlist</b> · <b>GlockApps</b> (inbox tests)                                                                             | Warm-up, rotating inboxes, A/B tests, LinkedIn steps: more emails reach the inbox and get replies                                              | \$          |

### Start free, add one at a time

Add a service only where a measured gap costs deals. Trial it for 4–6 weeks; keep it only if leads, replies or RFQs rise.

### Suggested order

Email verification, then contact data, then trade data, then a project database. Intent data last, after a coverage test.

### Cost bands (indicative)

\$ under \$100 a month · \$\$ \$100–1,000 · \$\$\$ over \$1,000. Many are annual contracts; get quotes.

## 10. Roadmap: work packages

The build is split into six work packages. Each ends with something the sales team can use and a measurable acceptance test.

**Timeline:** to be agreed with the client.

| Work package              | What it delivers                                                                                                                                                                                                                          | Accepted when                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 0. Secure and stabilise   | Login with Microsoft or Google accounts, roles, locked-down APIs, rotated keys, one database schema, contact bugs fixed, a real dashboard, and a test set of about 150 known contract awards                                              | No API responds without login; the dashboard matches the database; baseline accuracy measured                                                 |
| 1. Discovery agent        | Background agent runs, scheduled watchlists per country and product, tender-portal and project-news readers, quote-backed extraction, duplicate merging, new-lead alerts                                                                  | 85% or more of saved leads correct on the test set; 40 or more new qualified project or contractor accounts a week on the client's watchlists |
| 2. Contacts and CRM       | In-house contact finder, email verification, CRM screens or sync to the client's existing CRM, lead lists, export                                                                                                                         | 70% or more of priority accounts have at least one verified decision maker; estimated bounce rate under 3%                                    |
| 3. Validation agent       | Volza integration where available, product-to-customs-code mapping (e.g. 7304, 7305 and 7306 for steel pipe), purchase-history matching, need score with written reasons                                                                  | Every priority account shows a need score with sources; the client's sales lead agrees with 80% or more of the top-50 rankings                |
| 4. Outreach agent         | Mailbox connection, sending domains with SPF, DKIM and DMARC, warm-up, sequences, AI-drafted emails with human approval, reply sorting (interested, not now, wrong person, out of office, unsubscribe), CRM auto-update, opt-out register | Bounce under 2%, spam complaints under 0.3%, every reply sorted and logged within 5 minutes                                                   |
| 5. Hardening and handover | Monitoring, cost dashboard, audit log, backups, data-protection documentation, load testing, training, runbooks                                                                                                                           | A period of unattended operation with no critical incident                                                                                    |

**Scope options:** a lean scope covers work packages 0–2, with a simpler validation step and outreach drafted by AI but sent by people. An extended scope adds more countries, local-language sources and tender portals, an analytics warehouse, scoring that learns from won and lost deals, and a mobile view.

## 11. How it changes over time, and benchmarks

The system starts rule-heavy, with people reviewing every lead. It gets better every month as the test set grows and won and lost deals retune the rules. We measure it against published benchmarks and against what the existing tools deliver.

| Stage                    | What the system does                                                                 | People's role                       | Targets we measure against                                                               |
|--------------------------|--------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------|
| 1. Foundation            | Sources and rules live; AI reads pages; buyer graph fills up                         | Review every lead                   | 85% or more of saved leads correct on the test set                                       |
| 2. Contacts and outreach | Contact finder and outreach with approval; Volza checks if the client has an account | Approve every email                 | Verified contact on 50% or more of priority accounts; bounce rate under 2%               |
| 3. Learning              | Scores tuned on outcomes; weak sources dropped; repeated AI judgements become rules  | Approve new rules; spot-check leads | 90% or more correct; reply rate at or above the 3.43% cold-email average                 |
| 4. Scale                 | More countries, local-language sources and portals; auto-send on proven campaigns    | Handle RFQs and quotes              | Verified contact on 70% or more of priority accounts; AI cost per lead falling over time |

Targets are what we measure against, not guarantees. Work package 0 sets the real starting baseline.

## Compared with the existing tools

| Measure                                       | 6sense, ZoomInfo, Bombora                                                         | Our platform                                                         |
|-----------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Tenders, awards, bills of quantities, imports | Not covered                                                                       | Core signals                                                         |
| Evidence behind a lead                        | A topic score or a database record                                                | Source link and the exact quote                                      |
| Contact coverage in the client's markets      | Strongest for large enterprises                                                   | Senior contacts from public sources at first; grows with every reply |
| Outreach                                      | Separate tools                                                                    | Built in, from the client's mailbox                                  |
| Ownership                                     | Licensed; access ends with the subscription                                       | The client owns the data, rules and code                             |
| Yearly subscription                           | About \$113k–126k for all three ( <a href="#">Vendr</a> medians and entry prices) | None; only tools and AI usage (section 14)                           |

## External benchmarks we hold ourselves to

- **Reply rate:** the average cold-email reply rate in 2025 was 3.43%, and the top quarter of senders got 5.5% or more ([Instantly 2026 benchmark report](#)).
- **Deliverability:** Google's sender rules require SPF, DKIM and DMARC, one-click unsubscribe, and a spam rate below 0.30%, ideally below 0.10% ([Google](#)). Yahoo also requires unsubscribes to be honoured within 2 days ([Yahoo](#)).

## 12. Is it practical?

Yes for finding projects, buyers and senior contacts from free sources. Only partly for exact material quantities and verified emails, and not at all for company-level import data without the client's Volza account. The table is the honest picture.

| Capability                                                         | Possible with free sources and in-house code? | Notes                                                                                                               |
|--------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Find tenders, projects and contract awards in the client's markets | Yes                                           | Public portals, owner newsrooms, exchange announcements, trade press                                                |
| Name the EPC contractors                                           | Yes                                           | Awards are usually announced                                                                                        |
| Name subcontractors                                                | Partly                                        | Published less often; hiring and vendor-registration signals fill gaps                                              |
| Exact material quantities (bills of quantities)                    | Partly                                        | Inside tender booklets; needs the client's supplier registration, and some booklets carry a fee                     |
| Senior decision makers (directors, heads of procurement)           | Yes                                           | Company sites, press releases, annual reports                                                                       |
| Mid-level buyers and package engineers                             | Partly                                        | Less public; found through tender contacts and replies over time                                                    |
| Verified work emails                                               | Partly                                        | Email format plus a mail-server check works for many companies; domains that accept every address can't be verified |
| Company-level import history                                       | Only with Volza                               | No free company-level source in most countries; UN Comtrade gives country totals only                               |
| Web-research intent (Bombora-style)                                | No                                            | Not needed at the start; procurement signals are stronger here                                                      |
| Identify companies visiting the client's website                   | Partly                                        | Works for office networks; most mobile traffic can't be identified                                                  |

| Capability                                   | Possible with free sources and in-house code? | Notes                                                                         |
|----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------|
| Automated outreach from the client's mailbox | Yes                                           | Within daily sending limits, opt-out on every email and data-protection rules |

### What we need from the client

- A product list with customs (HS) codes, target countries and excluded accounts (current customers, competitors)
- Supplier registrations on the key tender portals, or permission to register on their behalf
- Login access to the tools they already pay for, such as Volza and their CRM
- Access to a mailbox and a separate sending domain for outreach, plus a named approver for emails
- A sales lead for about two hours a week to review leads and mark wins and losses

## 13. Development requirements

Building this well needs nine specialities working together. Each one covers a different angle of the product, so the client can see that every part of the system is owned.

| Speciality                         | What it covers in this project                                                                                                                                     |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI engineering                     | Agent design, prompts and model choice (Claude Haiku, Sonnet, Opus); quote-backed extraction; the test set of known contract awards; AI cost controls              |
| Data engineering and integrations  | Readers for tender portals, stock-exchange announcements and news feeds; company matching and de-duplication; the buyer graph; plug-ins for Volza and premium data |
| Full-stack product engineering     | Web app, review and approval screens, CRM screens or sync, dashboards, login and roles                                                                             |
| Email deliverability and outreach  | Gmail and Microsoft 365 APIs, sending domains with SPF, DKIM and DMARC, warm-up, sending limits, reply sorting, opt-out handling                                   |
| Security and compliance            | Access control, key management, audit log, data-protection rules in each market (e.g. UAE PDPL, EU GDPR), sanctions screening                                      |
| Cloud and operations               | Server, database, job queue, monitoring, backups, cost tracking                                                                                                    |
| QA and data quality                | Labelling the test set, accuracy reviews, regression tests before every release                                                                                    |
| Procurement and industry knowledge | How pipeline and industrial projects are bought: tenders, EPC awards, bills of quantities, customs codes; shaping the customer rules with the sales team           |
| Product design                     | Simple review and approval flows for the sales team; clear evidence and dashboards                                                                                 |

### Tools and environments needed to develop

- AI coding tools: Claude Code, and optionally Codex
- Code hosting and automated checks (GitHub)
- Separate development and production environments: database projects, a small server, a test mailbox and a test sending domain
- Claude API access with a spending cap
- The client inputs listed in section 12: product list with customs codes, portal registrations, tool logins, a mailbox and sending domain, and a sales lead to review results

## 14. Development and running cost: tools and APIs only

Once live, the system costs about \$130–365 a month to run, mostly AI usage and hosting. During the build, each developer needs an AI coding subscription of \$20–200 a month. Everything else uses free tiers, open-source

software, or tools the client already has.

**Tools needed during the build (per developer, per month)**

| Tool                         | What it is for                                          | Plan                                             | USD/month       |
|------------------------------|---------------------------------------------------------|--------------------------------------------------|-----------------|
| <a href="#">Claude Code</a>  | AI coding agent for building, testing and refactoring   | Pro \$20; Max \$100 or \$200 for heavy daily use | 20–200          |
| <a href="#">OpenAI Codex</a> | Second AI coding agent, e.g. for code review (optional) | Included in ChatGPT Plus \$20 or Pro \$100–200   | 0–200           |
| GitHub                       | Code hosting, reviews, automated checks                 | Free plan                                        | 0               |
| Supabase (development)       | Test database                                           | Free plan, 2 projects                            | 0               |
| Claude API (testing)         | Running the test set and trial runs                     | Pay per use                                      | 30–100 per team |

Example: three developers on Claude Max at \$100, plus one ChatGPT Plus seat for Codex reviews, cost about \$320 a month during the build, plus test API usage.

**Running cost once live (per month)**

| Item                            | Choice                                                                                                                                                                             | USD/month       |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| AI models                       | Claude Haiku 4.5 (\$1/\$5 per million input/output tokens) for filtering, Sonnet 5 (\$2/\$10) for extraction, Opus 5 (\$5/\$25) for qualification; Batch API halves scheduled runs | 100–300         |
| Database                        | <a href="#">Supabase Pro</a> (the free plan pauses after 7 days idle, so it is not for production)                                                                                 | 25              |
| Server for crawlers and workers | <a href="#">Hetzner</a> CX23 (€5.49) or a slightly larger machine                                                                                                                  | 7–20            |
| App hosting                     | The same server, or <a href="#">Vercel Pro</a> (the free Hobby plan forbids commercial use)                                                                                        | 0–20            |
| Web search and page reading     | GDELT, RSS feeds, self-hosted Crawl4AI, Brave within its free credit                                                                                                               | 0               |
| Contacts and email checks       | In-house finder, self-hosted Reacher, Hunter free plan                                                                                                                             | 0               |
| Email sending, trade data, CRM  | The client's existing email, Volza and CRM, or Zoho CRM Free                                                                                                                       | 0 new           |
| Monitoring                      | <a href="#">Sentry</a> free plan                                                                                                                                                   | 0               |
| <b>Total</b>                    |                                                                                                                                                                                    | <b>~130–365</b> |

Assumes about 2,000 accounts researched and 3,000 emails a month; AI cost grows with volume. Occasional one-off costs: tender booklets for priority tenders, and a sending domain (about \$10–15 a year). Prices checked in September 2026.

# What AI costs, and how it is billed

Two different kinds of AI cost. Neither includes premium data services or developer salaries.

## DURING THE BUILD

### AI for development

Fixed monthly subscription per developer. Stops or drops after the build.

| Tool                    | Plan                                   | USD / month    |
|-------------------------|----------------------------------------|----------------|
| Claude Code             | Claude Pro, or Max for heavy daily use | 20 · 100 · 200 |
| OpenAI Codex (optional) | Included in ChatGPT Plus or Pro        | 20 · 100–200   |
| Claude API for testing  | Pay per use, whole team                | 30–100         |

Example: 3 developers on Claude Max at \$100, plus one ChatGPT Plus seat for reviews, is about **\$320 a month**, plus test usage.

## ONCE LIVE

### AI for analysis inside the product

Claude API, pay per use: billed per million tokens read (input) and written (output).

| Model            | Used for                              | Input / output, per million tokens |
|------------------|---------------------------------------|------------------------------------|
| Claude Haiku 4.5 | Filtering pages                       | \$1 / \$5                          |
| Claude Sonnet 5  | Extracting companies, drafting emails | \$2 / \$10                         |
| Claude Opus 5    | Qualifying buyers                     | \$5 / \$25                         |

Scheduled runs through the Batch API cost half. At about 2,000 companies researched and 3,000 emails a month: **about \$100–300 a month**, or roughly **\$0.05–0.15 per company**.

### Total running cost once live: about \$130–365 a month

Claude API \$100–300, database \$25, server \$7–20, app hosting \$0–20. Everything else uses free tiers or the client's existing tools.

### What changes the bill

Countries and sources watched, pages read, emails drafted, and the model mix. Double the volume, roughly double the API cost.

### How we keep it under control

A monthly spending cap and alerts on the API account, a budget per run in our code, and cost per lead on the dashboard.

Disclaimer: vendor list prices were checked in September 2026 and exclude tax. Vendors can change prices, plan limits and usage allowances at any time. All figures are planning estimates, confirmed after the baseline from work package 0.

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## Sources

Checked in September 2026. Free tools and hosting: [GDELT](#), [Brave Search API](#), [Google Custom Search](#), [Crawl4AI](#), [Crawlee](#), [Jina Reader](#), [Reacher](#), [Hunter](#), [Leadfeeder](#), [Zoho CRM](#), [HubSpot CRM](#), [Supabase billing](#), [Supabase pausing](#), [Supabase Queues](#), [Vercel fair use](#), [Hetzner](#), [Sentry](#), [UN Comtrade](#), [Gemini API terms](#). AI tools: [Claude plans](#), [ChatGPT and Codex plans](#), [Claude Batch API](#). Market data and law: [Tadawul announcement example](#), [Tadawul disclosure rules](#), [OFAC SDN](#), [UN sanctions list](#), [UAE local list](#), [UAE PDPL](#), [Saudi PDPL](#), [Etihad regulations](#). Existing tools and benchmarks: [6sense on Vendr](#), [ZoomInfo on Vendr](#), [Bombora pricing](#), [Bombora co-op](#), [6sense intent data](#), [Instantly benchmark report](#), [Google sender rules](#), [Yahoo sender rules](#), [AWS](#), [Cloudflare](#), [Fastly](#). Claude model prices are from Anthropic's API rate card. Code findings are from our audit of the prototype repository on 24 Sep 2026.